CS728 Programming Assignment 2

March 12, 2026

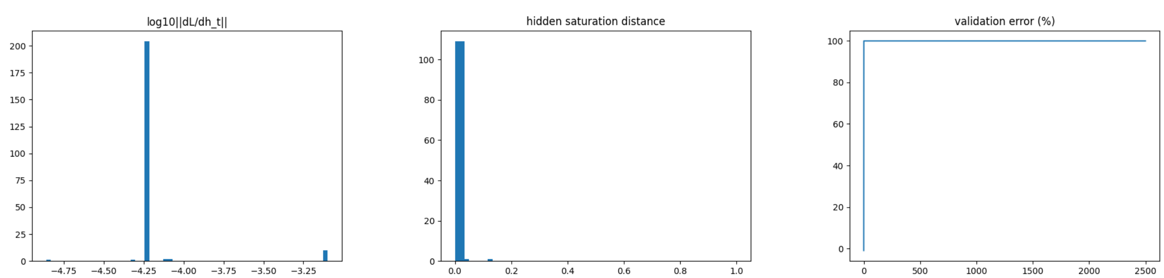
1 Summary Table

Run	Task	Model	Clip	$\log_{10} \|dL/dh_t\|$	μ_{sat}	$sat < 0.05(\%)$	$\rho(W_{hh})$ (last)	bestValidErr(%)
A1	mem	RNN	none	-4.22	0.017	99.5	14.20	100.00
A2	mem	RNN	0.05	-4.58	0.628	0.0	1.14	100.00
A3	mem	RNN	0.01	-4.60	0.611	0.0	1.12	100.00
A4	mem	GRU	none	-6.07	0.538	0.0	2.01	100.00
A5	mem	GRU	0.05	-4.18	0.850	0.0	1.29	100.00
B1	mul	RNN	none	-12.00	0.939	0.0	0.71	25.90
B2	mul	GRU	none	-12.00	0.995	0.0	0.08	24.25

Table 1: Per Run Summary

2 Results: Memorization (A1–A5)

2.1 A1: RNN (tanh), no clipping



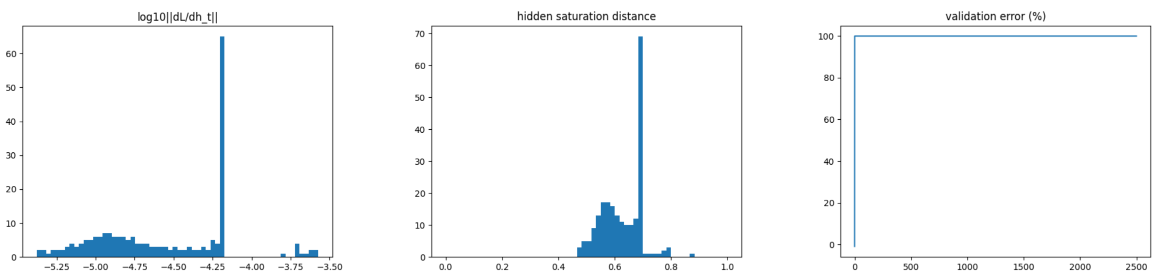
(a) $\log_{10} \|dL/dh_t\|$ (b) hidden sat distance (c) valid error

Figure 1: A1 diagnostics.

Q1 (vanish/explode?): $\log_{10} \|dL/dh_t\|$ is concentrated around ≈ -4.2, so the gradient-through-time signal is not extremely vanishing (not near -8 to -12) and not exploding.
Q2 (saturation?): hidden saturation distance mass is extremely close to 0 (Table **??**: 99.5% of time steps have sat-distance < 0.05), indicating strong tanh saturation. This saturation co-occurs with a very large $\rho(W_{hh})$ (final ≈ 14.2), suggesting unstable recurrent dynamics where saturation

can suppress useful learning signals even if g_t is not numerically tiny.
Validation: validation error remains at 100% throughout, so the model does not learn the memorization task in this regime.

2.2 A2: RNN (tanh), clipping cutoff 0.05

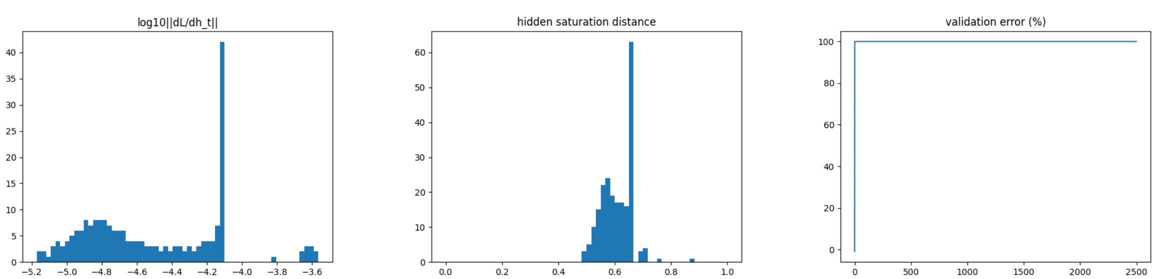


(a) $\log_{10} \|dL/dh_t\|$ (b) hidden sat distance (c) valid error

Figure 2: A2 diagnostics.

Q1: the histogram shifts slightly more negative than A1 (median ≈ -4.6), but it is still far from the extreme vanishing regime.
Q2: saturation largely disappears (mean sat-distance ≈ 0.63; sat$< 0.05\% = 0$), indicating hidden states mostly remain in the linear region of tanh. This shows that strong saturation is *not* necessary for failure on this task.
Validation: validation error stays at 100%.

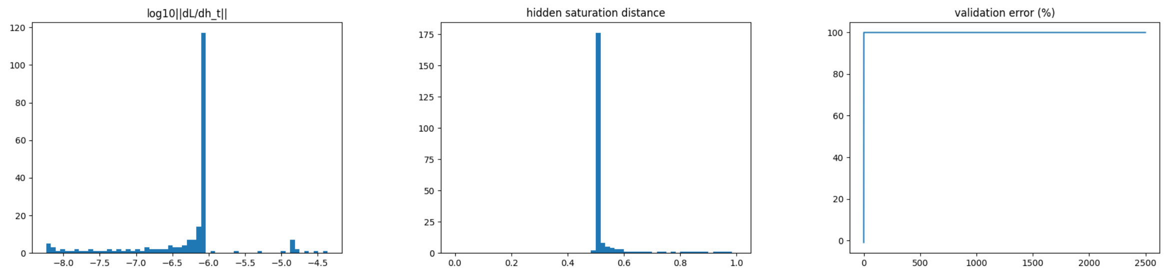
2.3 A3: RNN (tanh), clipping cutoff 0.01



(a) $\log_{10} \|dL/dh_t\|$ (b) hidden sat distance (c) valid error

Figure 3: A3 diagnostics.

Q1: similar to A2 (median ≈ -4.6), indicating neither extreme vanishing nor explosion in the g_t histogram.
Q2: similar non-saturated hidden behavior to A2 (mean sat-distance ≈ 0.61).
Validation: validation error stays at 100%.



(a) $\log_{10} \|dL/dh_t\|$ (b) hidden sat distance (c) valid error

Figure 4: A4 diagnostics.

2.4 A4: GRU, no clipping (gate diagnostics)

Q1: compared to the RNN runs, the GRU shows substantially more negative values (median ≈ -6.1 with a tail to ≈ -8), indicating stronger gradient vanishing across time.
Q2: hidden saturation distance is centered around ≈ 0.54 (not saturated). Therefore, vanishing gradients can occur even when tanh units are not close to ± 1.
Q4 (gate saturation): Figure 5 shows gate saturation distances. The update gate has mean $d(z_t) \approx 0.182$ (closer to saturation than 0.5), while the reset gate is largely unsaturated with mean $d(r_t) \approx 0.456$ (near 0.5). This suggests the GRU is often operating with a somewhat saturated update gate, but this alone does not prevent gradient vanishing in this regime.
Validation: validation error stays at 100%.

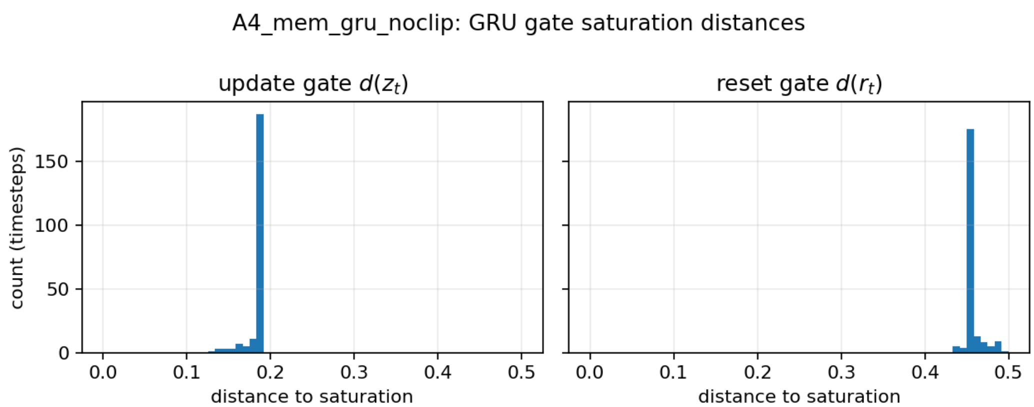
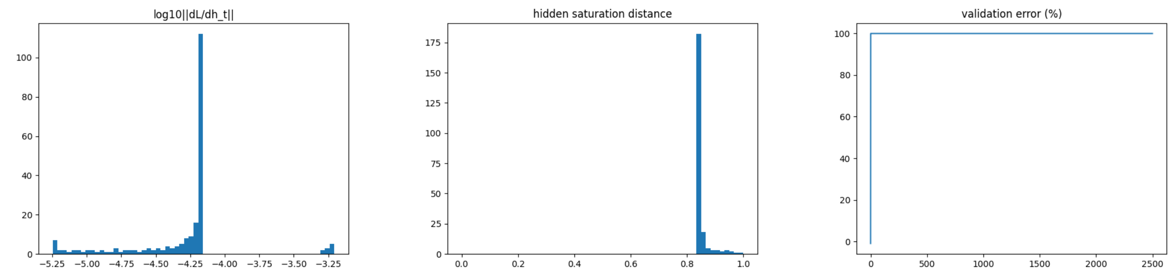


Figure 5: A4 GRU gate saturation-distance histograms.

2.5 A5: GRU, clipping cutoff 0.05 (gate diagnostics)

Q1: clipping shifts the GRU g_t histogram towards larger values (median ≈ -4.2), i.e. much less vanishing than A4.
Q2: hidden saturation distance moves close to 1 (mean ≈ 0.85), indicating hidden states are very close to 0 (strongly unsaturated).
Q4 (gate saturation): Figure 7 shows that the update gate becomes even more saturated (mean $d(z_t) \approx 0.125$), while the reset gate remains mostly unsaturated (mean $d(r_t) \approx 0.492$).
Validation: validation error stays at 100% despite improved gradient-through-time diagnostics.



(a) $\log_{10} \|dL/dh_t\|$ (b) hidden sat distance (c) valid error

Figure 6: A5 diagnostics.

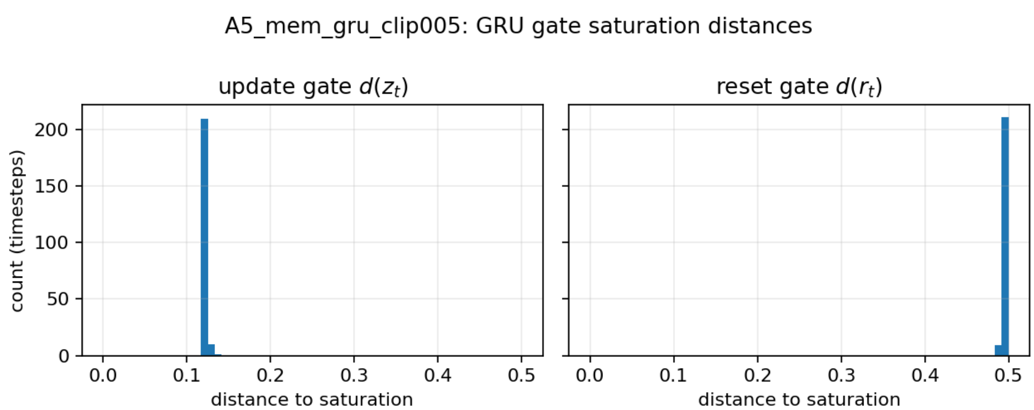
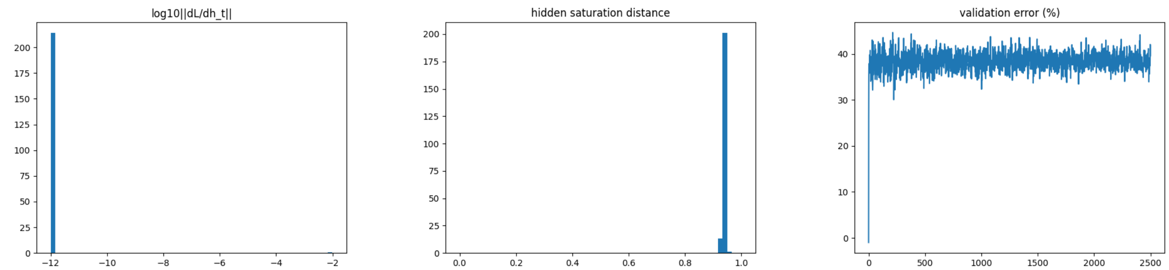


Figure 7: A5 GRU gate saturation-distance histograms.

3 Results: Multiplication (B1–B2)

3.1 B1: RNN (tanh), no clipping

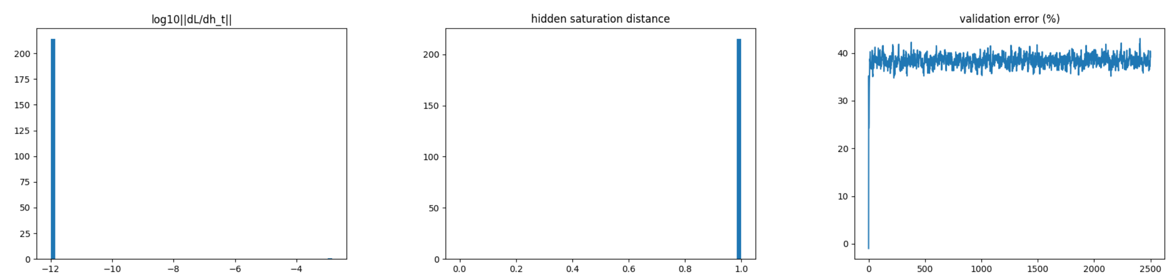


(a) $\log_{10} \|dL/dh_t\|$ (b) hidden sat distance (c) valid error

Figure 8: B1 diagnostics.

Q1: the histogram is essentially a spike at $\log_{10} \|dL/dh_t\| \approx -12$, indicating extreme vanishing gradient-through-time.
Q2: hidden saturation distance is near 0.94 (i.e. $|h| \approx 0.06$), so the hidden state is *not* saturated; instead it is very close to 0 (linear region). This is a clear counterexample where vanishing gradients occur without tanh saturation.
Validation: validation error fluctuates around $\approx$ 38–42% (best $\approx$ 25.9%), indicating limited learning progress in this regime.

3.2 B2: GRU, no clipping (gate diagnostics)



(a) $\log_{10} \|dL/dh_t\|$ (b) hidden sat distance (c) valid error

Figure 9: B2 diagnostics.

Q1: like B1, the gradient-through-time signal is extremely vanishing (median ≈ -12).
Q2: hidden saturation distance is even closer to 1 (mean $\approx$ 0.995), so hidden activations are very close to 0.
Q4 (gate saturation): Figure 10 shows $d(z_t) \approx 0.119$ (update gate more saturated) and $d(r_t) \approx 0.499$ (reset gate almost perfectly unsaturated). Despite gating, the long-range regression signal still collapses over time in this configuration.
Validation: validation error is similar to B1 (best $\approx$ 24.25%; final $\approx$ 39%).

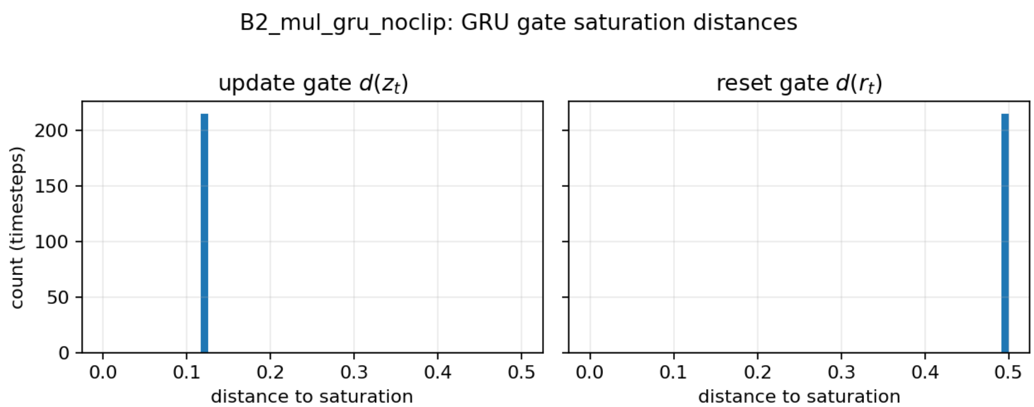


Figure 10: B2 GRU gate saturation-distance histograms.

4 Comparisons (Q3–Q5)

Q3: No clipping vs clipping

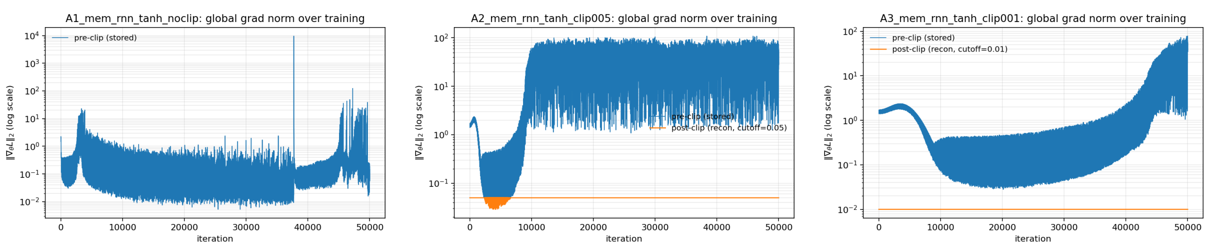
RNN on memorization (A1 vs A2/A3):

- **Gradient norm:** A2 shows very large pre-clip global norms (median ≈ 40.8), implying parameter-gradient explosion that is subsequently clipped (post-clip norm fixed at the cutoff). In contrast, A1 has modest median grad norms but very large $\rho(W_{hh})$ (Figure 13), indicating unstable recurrent dynamics and saturation.

- $\log_{10}(g_t)$ **histogram:** clipping slightly shifts the distribution (A2/A3 more mass around ≈ -4.6), but none of A1–A3 reaches the extreme vanishing regime.

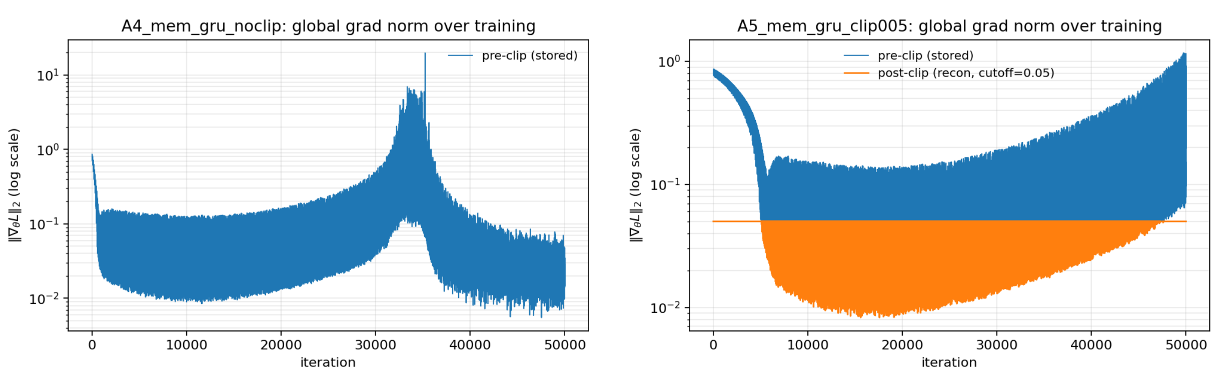
- **Saturation:** clipping removes strong tanh saturation (A1: sat-distance near 0; A2/A3: sat-distance around 0.6).

GRU on memorization (A4 vs A5): clipping moves g_t from clearly vanishing (A4 median ≈ -6.1) to much healthier values (A5 median ≈ -4.2), and makes hidden activations strongly unsaturated (A5 sat-distance ≈ 0.85). Despite these improvements, validation error remains 100% for both.



(a) A1 (no clip) (b) A2 (cutoff 0.05) (c) A3 (cutoff 0.01)

Figure 11: Memorization RNN: global gradient norm over training (log scale). For clipped runs, the reconstructed post-clip norm (orange) is capped at the cutoff.



(a) A4 (no clip) (b) A5 (cutoff 0.05)

Figure 12: Memorization GRU: global gradient norm over training (log scale).

Q4: RNN vs GRU (including gates)

- **Memorization:** the unclipped GRU (A4) exhibits stronger gradient vanishing than the unclipped RNN (A1), even though the RNN has more hidden saturation. Gate diagnostics show the reset gate is largely unsaturated ($d(r_t) \approx 0.46$–0.49), while the update gate is more saturated ($d(z_t) \approx 0.12$–0.18).
- **Multiplication:** both RNN and GRU collapse to extreme vanishing ($\log_{10}(g_t) \approx -12$) while remaining unsaturated, so the dominant failure mode here is not tanh saturation but the long-range gradient decay inherent to the task and this configuration.
- **GRU-specific failure modes (observed):** the update gate tends to be saturated (low $d(z_t)$), which can reduce effective state updates and make it hard to route precise long-range information; in B2, gating does not prevent vanishing gradients for the long-range regression signal.

Q5: $\rho(W_{hh})$ over training and correlation

Figure 13 and Figure 14 summarize $\rho(W_{hh})$:

- In A1, $\rho(W_{hh})$ grows very large (≈ 14) and the network becomes heavily saturated; this correlates with unstable dynamics and failure to learn.
- With clipping (A2/A3/A5), $\rho(W_{hh})$ stays close to 1–1.3, consistent with stabilizing updates and reducing saturation/explosion risk.
- In B1/B2, $\rho(W_{hh}) < 1$ (especially B2 where $\rho \approx 0.08$) indicates strongly contractive recurrence; this aligns with the extreme vanishing gradients seen in the $\log_{10}(g_t)$ histograms.

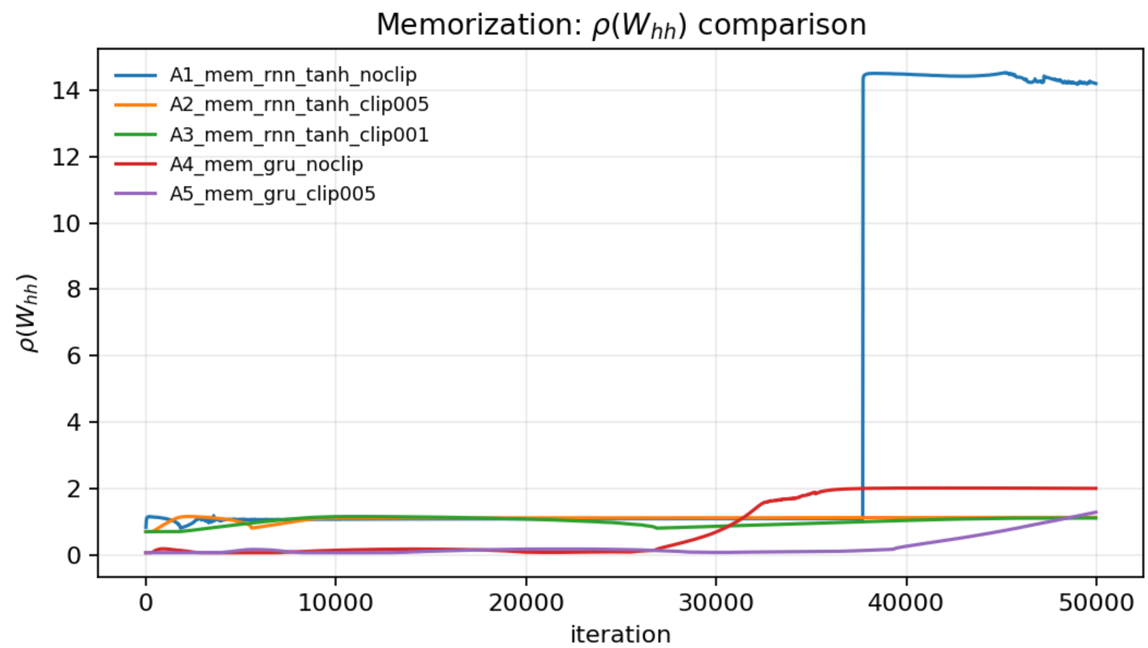


Figure 13: Memorization runs: $\rho(W_{hh})$ over training (all A1–A5).

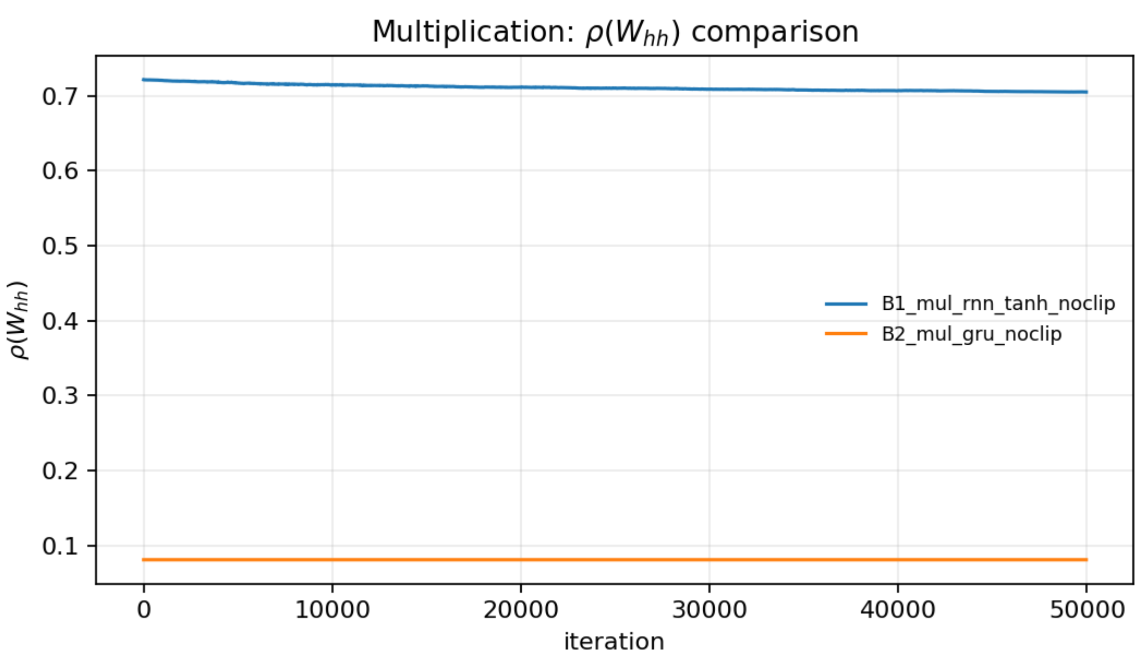


Figure 14: Multiplication runs: $\rho(W_{hh})$ over training (B1–B2).

A Commands (from assignment PDF)

The required commands (as provided in the PDF) are reproduced here for completeness:

```
# Common flags:
--nhid 50 --lr 0.01 --bs 20 --min_length 50 --max_length 200
--maxiters 50000 --ebs 10000 --cbs 1000 --checkFreq 20
--seed 52 --valid_seed 12345 --collectDiags --diagBins 60 --satThresh 0.05

# A1: mem, RNN tanh, no clipping
python -m trainingRNNs_torch.train --task mem --model rnn --alpha 0.0 \
  --clipstyle nothing --name A1_mem_rnn_tanh_noclip [common flags...]

# A2: mem, RNN tanh, clipping cutoff 0.05
python -m trainingRNNs_torch.train --task mem --model rnn --alpha 0.0 \
  --clipstyle rescale --cutoff 0.05 --name A2_mem_rnn_tanh_clip005 [common flags...]

# A3: mem, RNN tanh, clipping cutoff 0.01
python -m trainingRNNs_torch.train --task mem --model rnn --alpha 0.0 \
  --clipstyle rescale --cutoff 0.01 --name A3_mem_rnn_tanh_clip001 [common flags...]

# A4: mem, GRU, no clipping, with gate diagnostics
python -m trainingRNNs_torch.train --task mem --model gru --alpha 0.0 \
  --clipstyle nothing --diagGates --name A4_mem_gru_noclip [common flags...]

# A5: mem, GRU, clipping cutoff 0.05, with gate diagnostics
python -m trainingRNNs_torch.train --task mem --model gru --alpha 0.0 \
  --clipstyle rescale --cutoff 0.05 --diagGates --name A5_mem_gru_clip005 [common flags...]

# B1: mul, RNN tanh, no clipping
python -m trainingRNNs_torch.train --task mul --model rnn --alpha 0.0 \
  --clipstyle nothing --name B1_mul_rnn_tanh_noclip [common flags...]

# B2: mul, GRU, no clipping, with gate diagnostics
python -m trainingRNNs_torch.train --task mul --model gru --alpha 0.0 \
  --clipstyle nothing --diagGates --name B2_mul_gru_noclip [common flags...]
```

B Reproducibility Notes

All plots embedded in this report are stored in `report/images/`. Additional plots (gradient norm curves, $\rho(W_{hh})$ curves, and GRU gate histograms) were generated from the saved `*_final_state.npz` files using `report/generate_extra_plots.py`.